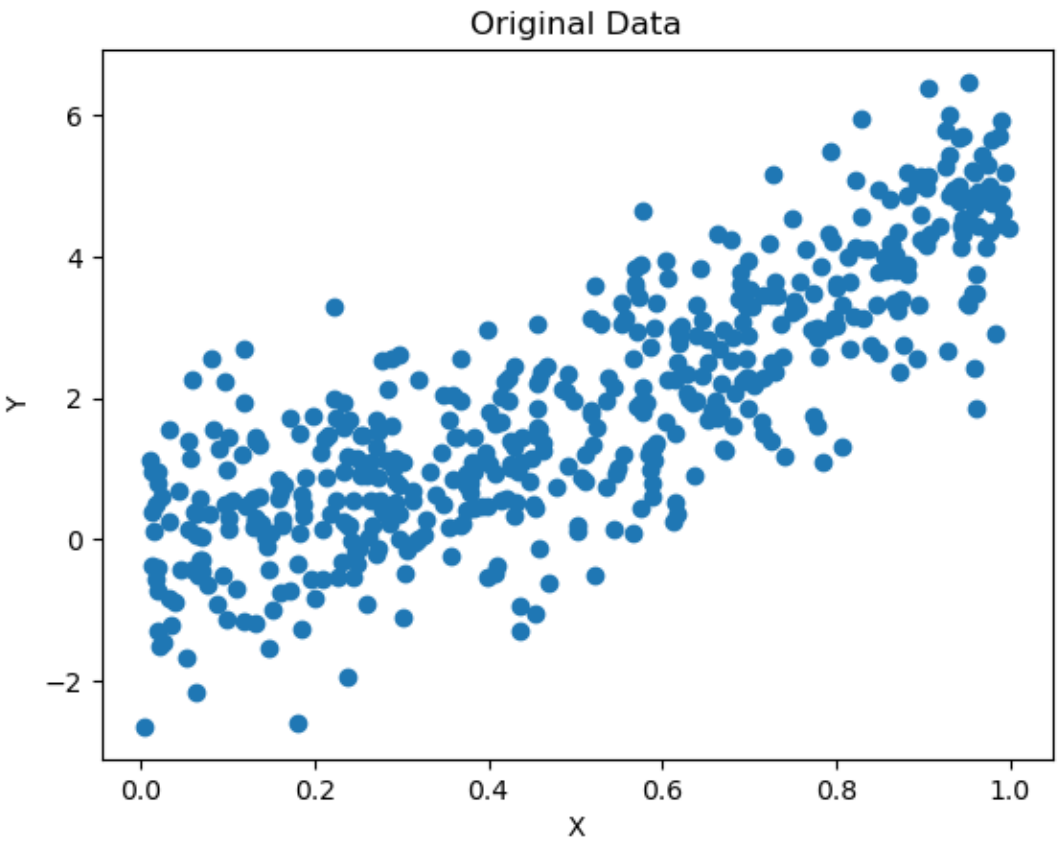
Prerequisite & Data Generation

```python
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats
import random
import typing
```

✓ 0.0s

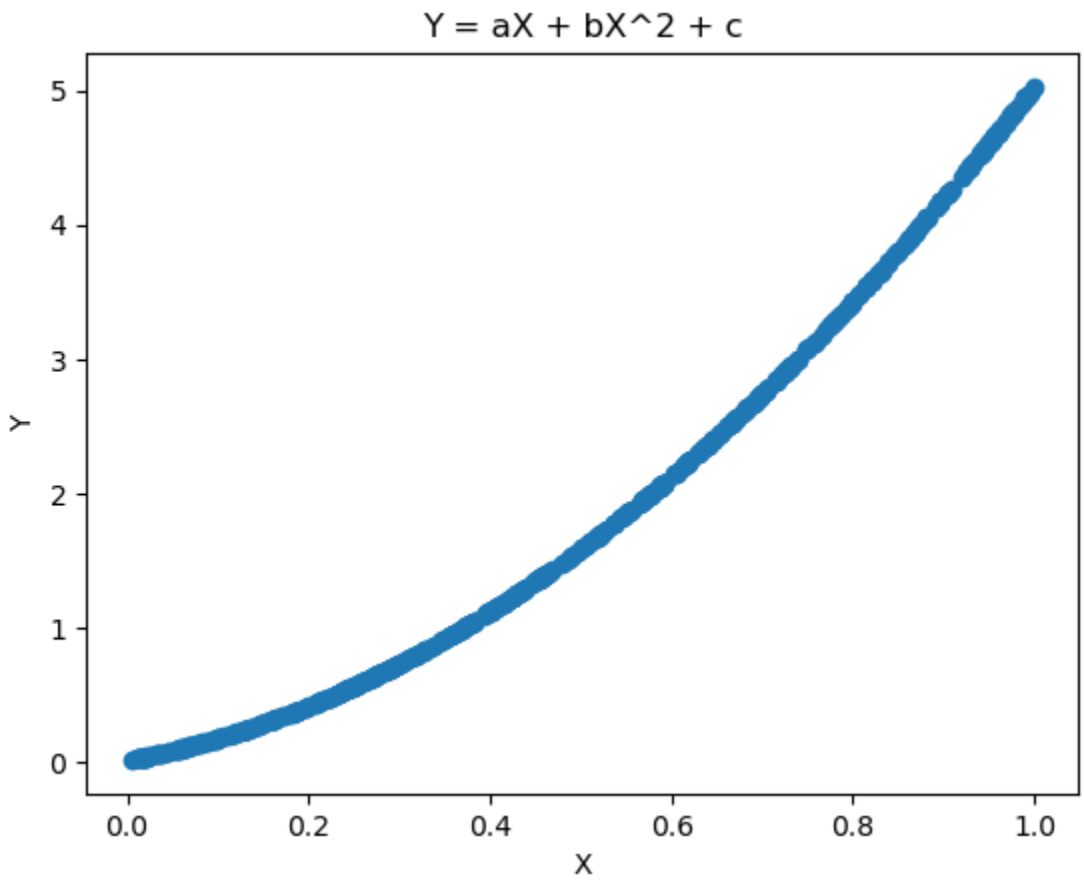
```python
# Data Generation
np.random.seed(0)
n = 500     # sample size
X = np.random.uniform(0, 1, n)
epsilon = np.random.normal(0, 1, n)
a, b, c = 2, 3, 1
Y = a * X + b * X ** 2 + epsilon
plt.scatter(X, Y)
plt.title("Original Figure")
plt.xlabel("X")
plt.ylabel("Y")
```

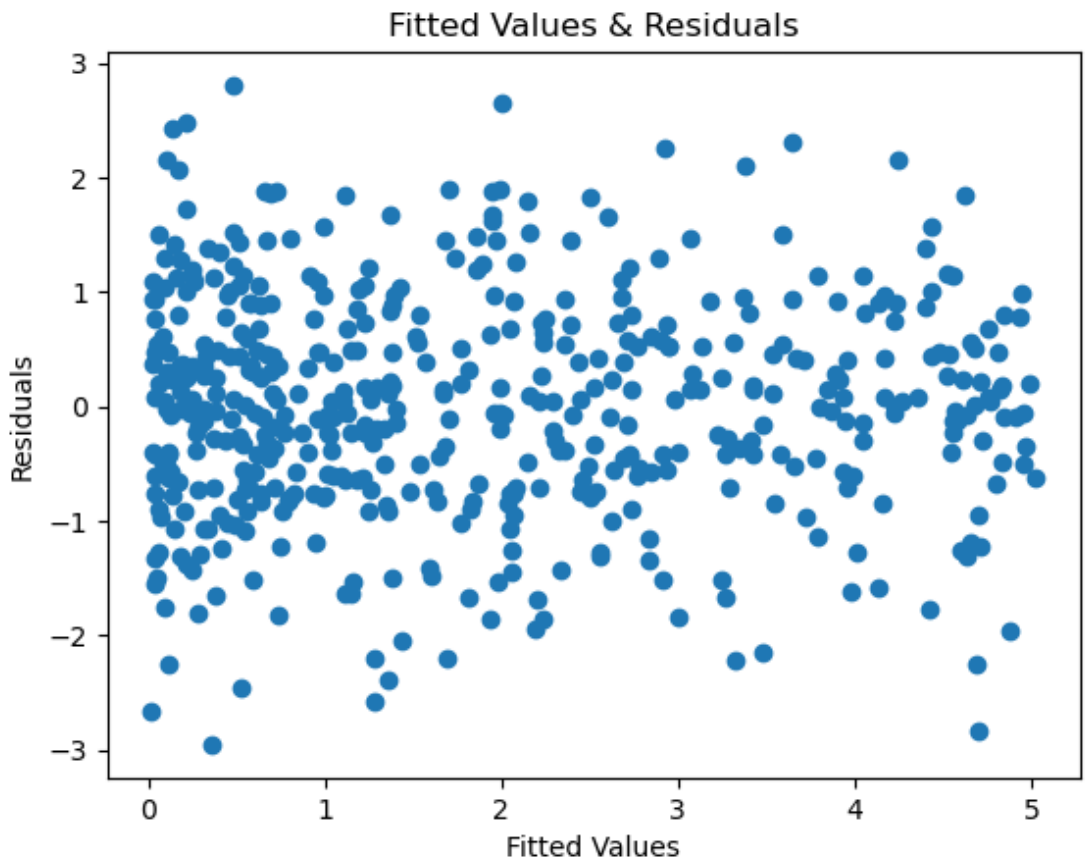
✓ 0.1s

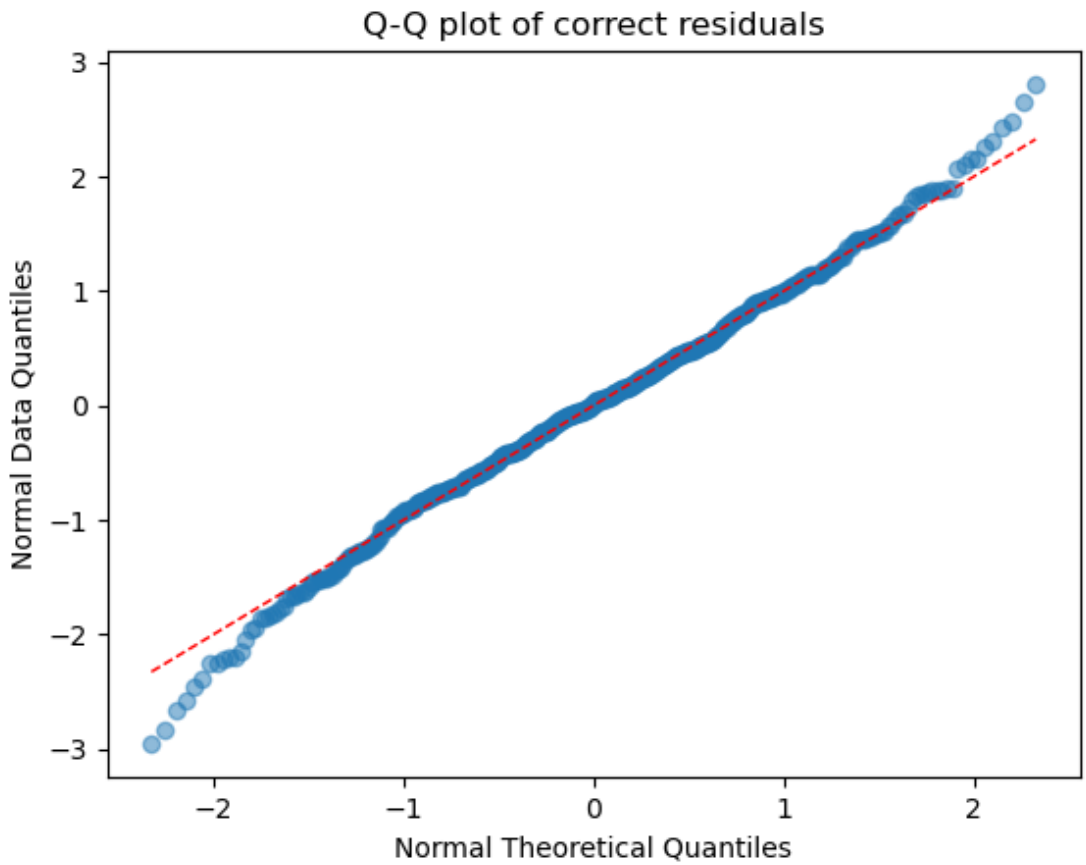


Task 1: Fit a Correct Linear Model

```python
# 设计矩阵: 1, X, X^2
X_design = np.column_stack((np.ones(n), X, X ** 2))
# 最小二乘解 beta = (X'X)^{-1} X'Y
XtX = X_design.T @ X_design
XtX_inv = np.linalg.inv(XtX)
beta = XtX_inv @ (X_design.T @ Y)

Y_fit = X_design @ beta
residuals = Y - Y_fit
res_sorted = np.sort(residuals)

# X, Y_fit
plt.figure()
plt.scatter(X, Y_fit)
plt.title("Y = aX + bX^2 + c")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()

# residuals, Y_fit
plt.figure()
plt.scatter(Y_fit, residuals)
plt.title("Fitted Values & Residuals")
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.show()

# Q-Q plot of residuals
quantiles = stats.norm.ppf(np.linspace(0.01, 0.99, n))  # 为避免无穷，取0.01~0.99
plt.figure()
plt.scatter(quantiles, res_sorted, alpha=0.5)
plt.plot([quantiles.min(), quantiles.max()], [quantiles.min(), quantiles.max()], 'r--', linewidth=1)
plt.xlabel('Normal Theoretical Quantiles')
plt.ylabel('Normal Data Quantiles')
plt.title('Q-Q plot of correct residuals')
plt.show()
```




Fitted Values & Residuals
Residuals
Fitted Values
3
2
1
0
−1
−2
−3
0
1
2
3
4
5


Q-Q plot of correct residuals
Normal Data Quantiles
Normal Theoretical Quantiles
3
2
1
0
−1
−2
−3
−2
−1
0
1
2

Discussion:

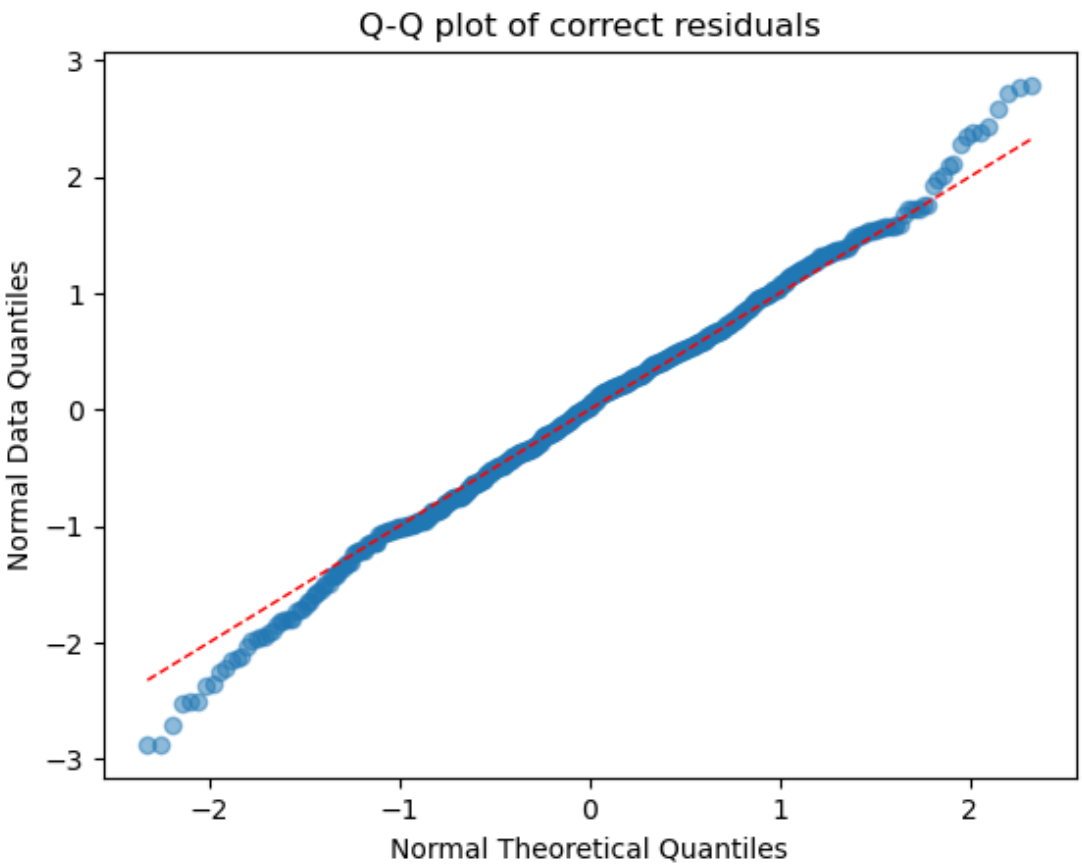
1. Pattern in the residual plot. Residuals are randomly distributed around the zero line.
2. Normality assumption. The normality assumption appears valid, as it is shown on the Q-Q plot, the points are located near the diagonal.

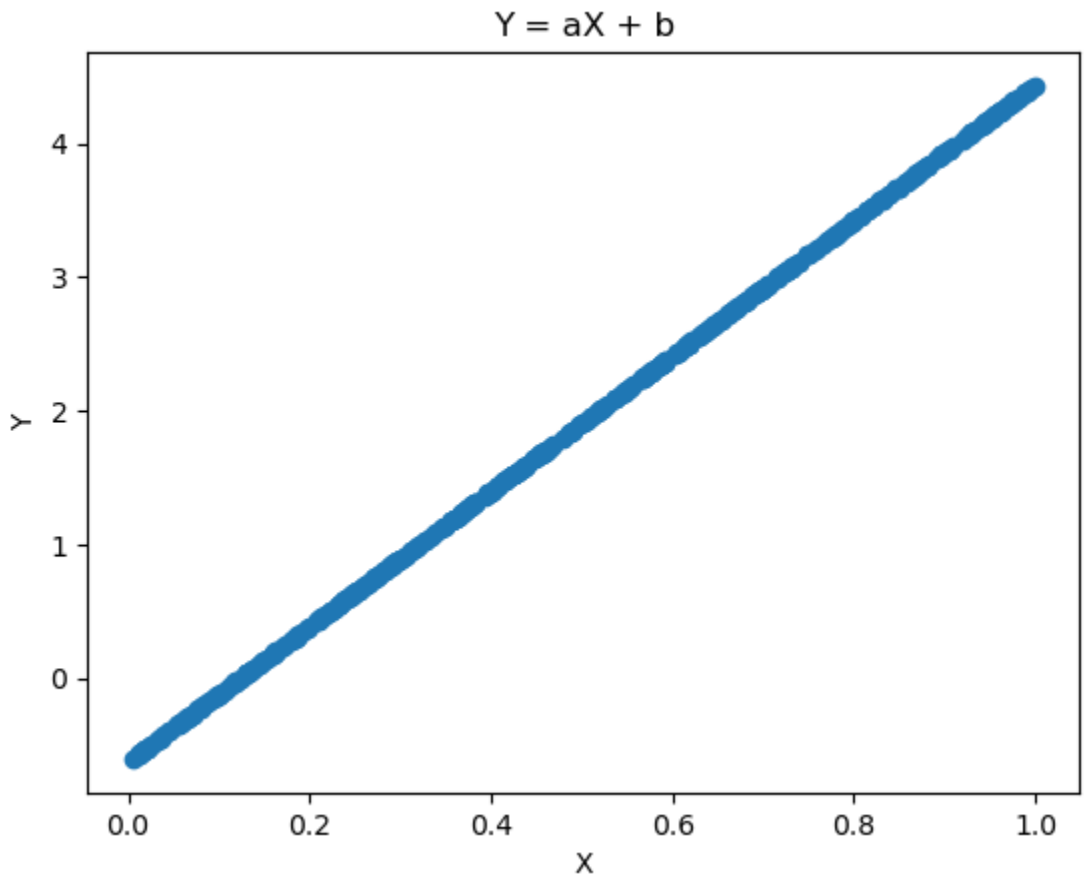
Task 2: Fit A Misspecified Model

```python
### Task 2

# parameters
a_hat = np.sum((X - np.mean(X)) * (Y - np.mean(Y))) / np.sum((X - np.mean(X)) ** 2)
b_hat = np.mean(Y) - a_hat * np.mean(X)

Y_fit = a_hat * X + b_hat
residuals = Y - Y_fit
res_sorted = np.sort(residuals)

# X, Y_fit
plt.figure()
plt.scatter(X, Y_fit)
plt.title("Y = aX + b")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()

# Y_fit, residuals
plt.scatter(Y_fit, residuals)
plt.title("Fitted Values & Residuals")
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.show()

# Q-Q plot of residuals
quantiles = stats.norm.ppf(np.linspace(0.01, 0.99, n))  # 为避免无穷，取0.01~0.99
plt.figure()
plt.scatter(quantiles, res_sorted, alpha=0.5)
plt.plot([quantiles.min(), quantiles.max()], [quantiles.min(), quantiles.max()], 'r--', linewidth=1)
plt.xlabel('Normal Theoretical Quantiles')
plt.ylabel('Normal Data Quantiles')
plt.title('Q-Q plot of correct residuals')
plt.show()
```

0.2s

Fitted Values & Residuals
Residuals
Fitted Values
3
2
1
0
−1
−2
−3
0
1
2
3
4


Q-Q plot of correct residuals
Normal Data Quantiles
Normal Theoretical Quantiles
3
2
1
0
−1
−2
−3
−2
−1
0
1
2

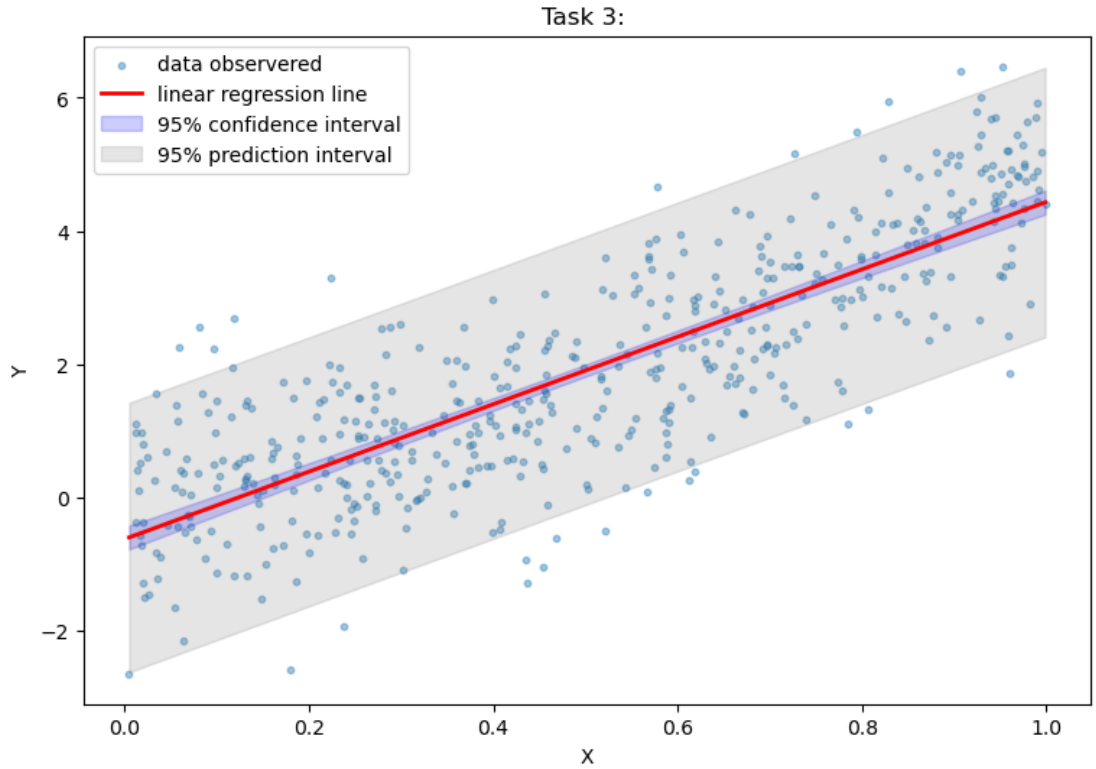


Comparison:

1. The model in task 1 fits better.
2. The normality assumption of residuals is not valid. It's obvious that there exists a clear deviation between the data points and the diagonal.
3. Pass

Task 3: Confidence Interval and Prediction Interval

```python

n = len(X)
x_bar = np.mean(X)
Sxx = np.sum((X - x_bar) ** 2)
sigma_hat = np.sqrt(np.sum(residuals ** 2) / (n - 2))

# t 临界值 (95% 置信水平, 双尾)
from scipy.stats import t
t_crit = t.ppf(0.975, df=n-2)

# 生成一组 X0 用于画图
X0 = np.linspace(min(X), max(X), 200)
y0_hat = a_hat * X0 + b_hat

# 均值的标准误
se_mean = sigma_hat * np.sqrt(1/n + (X0 - x_bar)**2 / Sxx)
# 预测的标准误 (包含个体误差)
se_pred = sigma_hat * np.sqrt(1 + 1/n + (X0 - x_bar)**2 / Sxx)

# 置信区间和预测区间
ci_lower = y0_hat - t_crit * se_mean
ci_upper = y0_hat + t_crit * se_mean
pi_lower = y0_hat - t_crit * se_pred
pi_upper = y0_hat + t_crit * se_pred

plt.figure(figsize=(9, 6))
plt.scatter(X, Y, alpha=0.4, s=10, label='data observered')
plt.plot(X0, y0_hat, 'r-', linewidth=2, label='linear regression line')
plt.fill_between(X0, ci_lower, ci_upper, color='blue', alpha=0.2, label='95% confidence interval')
plt.fill_between(X0, pi_lower, pi_upper, color='gray', alpha=0.2, label='95% prediction interval')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Task 3: ')
plt.legend()
plt.show()
```



 # Task 4: Discussion

1. How does model misspecification affect residual behavior?

Model misspecification makes residuals lose its random distribution around zero, and deviate from normality assumption.

2. How does it affect inference (confidence intervals)?

The standard error estimates are distorted and parameter point estimates are biased, thus, the confidence intervals have inaccurate width.

3. Why is it important to check diagnostic plots in regression?

Diagnostic plots clearly reveal violations of linearity and normality, helping us avoid wrong assumptions and misleading conclusions, and improves model specification.